

FEA HW1

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1 Taylor and Interpolation

1.1

$$f(x) = (x + \cos x)^2 \sin x$$

If the desired truncation error is $\mathcal{O}(h^4)$ then the approximating polynomial must be of order 3. This means we need 4 nodal points to interpolate with Lagrange polynomials, and 4-element vectors. Since the nodal points must be equally spaced in the interval,

$$\begin{aligned} f_i &= [f(0) \quad f(\frac{1}{6}) \quad f(\frac{1}{3}) \quad f(\frac{1}{2})]^T \\ \Rightarrow f_i &= [0 \quad 0.2205 \quad 0.5346 \quad 0.9098]^T \\ \Rightarrow N_i(x) &= \left[\begin{array}{cccc} \frac{(x-\frac{1}{6})(x-\frac{1}{3})(x-\frac{1}{2})}{-\frac{1}{36}} & \frac{x(x-\frac{1}{3})(x-\frac{1}{2})}{\frac{1}{108}} & \frac{x(x-\frac{1}{6})(x-\frac{1}{2})}{-\frac{1}{108}} & \frac{x(x-\frac{1}{6})(x-\frac{1}{3})}{\frac{1}{36}} \end{array} \right] \end{aligned}$$

1.2

See MATLAB file.

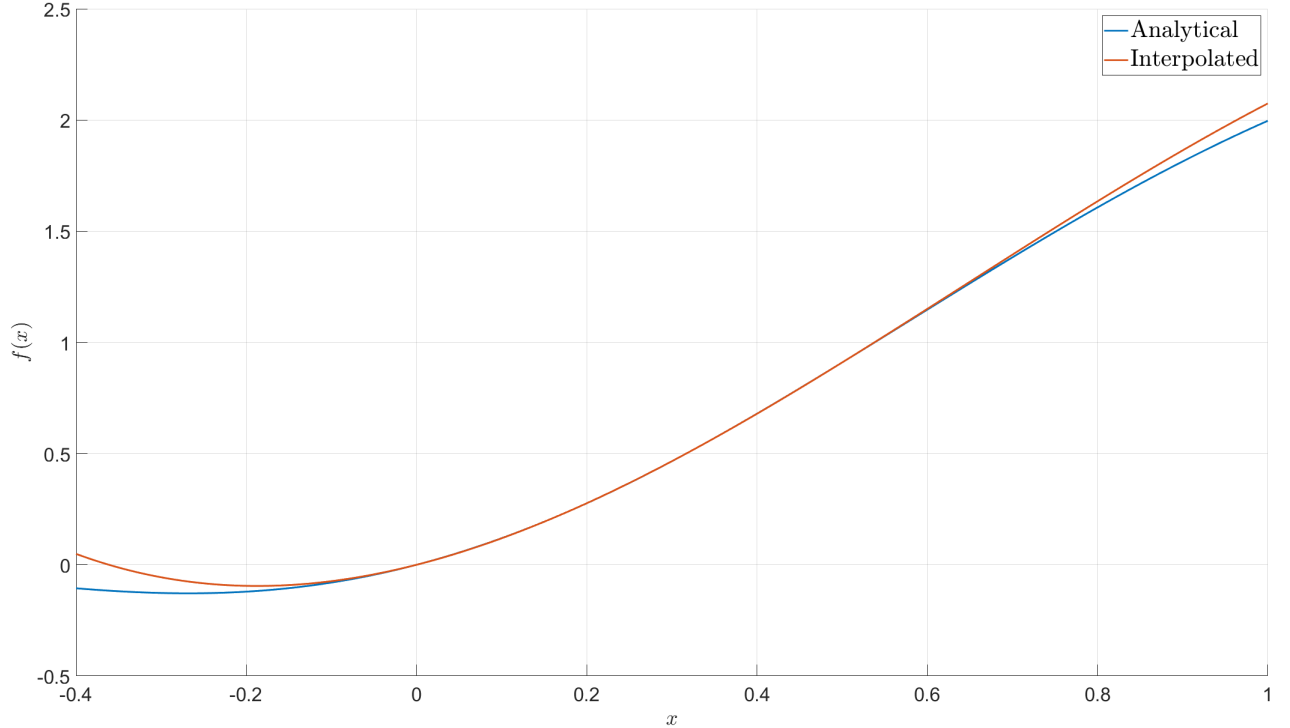


Figure 1: Q1a: $f(x)$ vs. Third order Interpolation of $f(x)$

The interpolation is exact at the equally spaced nodes as expected. The error is smallest in close proximity to each node, which is also expected. The error decays as proximity to any node decreases.

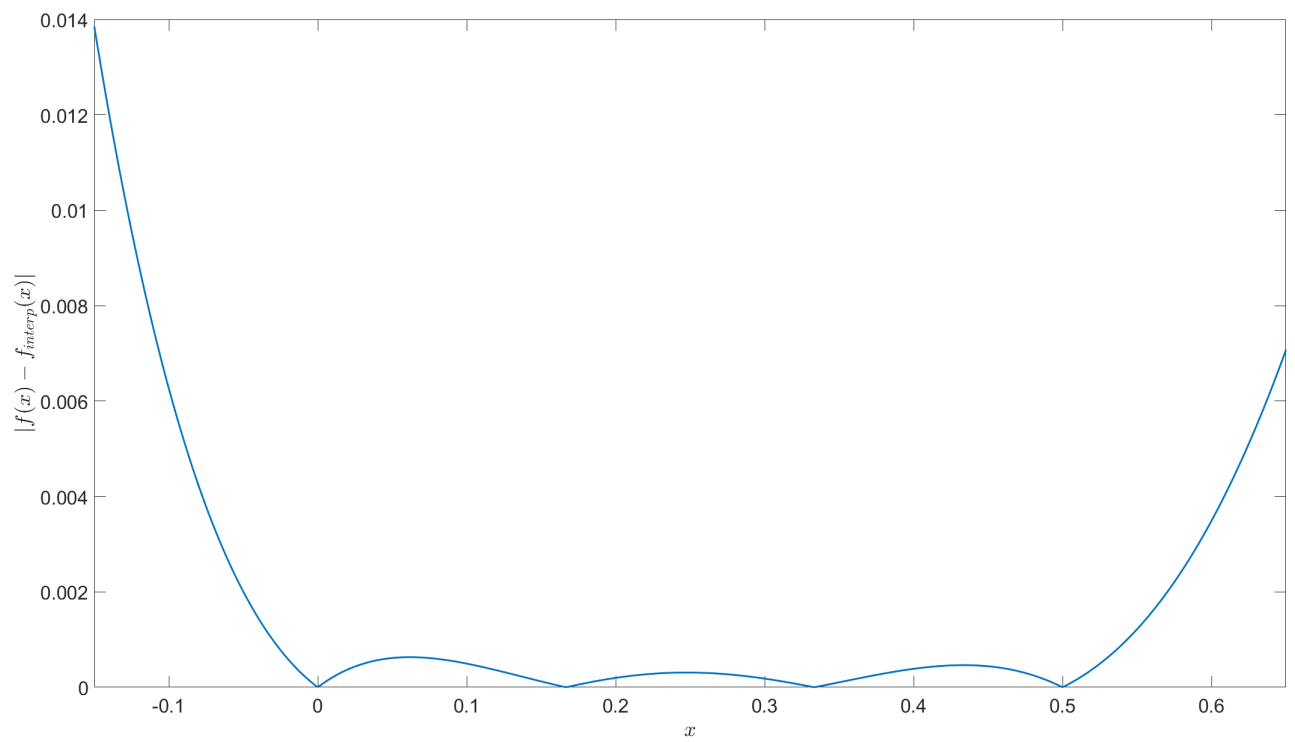


Figure 2: Q1b: Interpolation error

2 1D Integration

See MATLAB file.

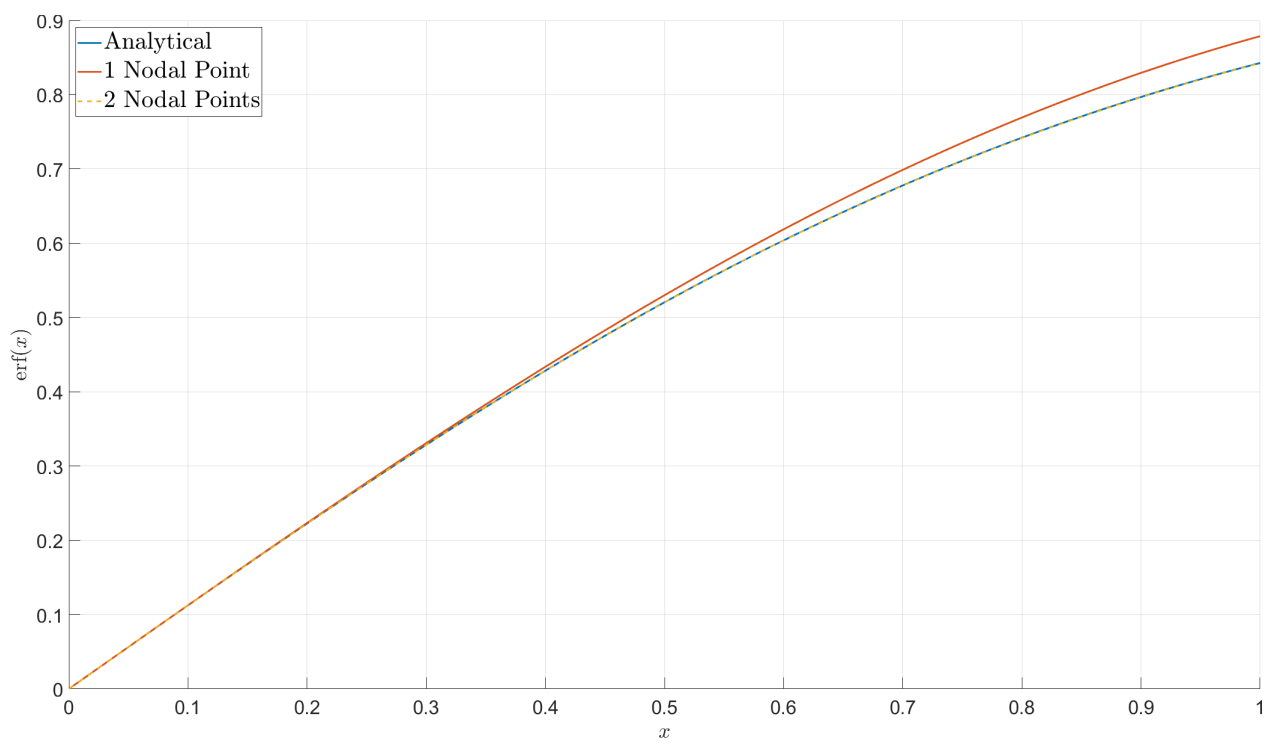


Figure 3: Q2: $\text{erf}(x)$ vs. Various Gauss Quadratures of $\text{erf}(x)$